

Experimental Design and Data Analysis: Assignment 2

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Exercise 1. Trees

a) We begin by generating the ANOVA table without taking diameter and height into account:

```
## Analysis of Variance Table
##
## Response: volume
##           Df Sum Sq Mean Sq F value Pr(>F)
## type       1    380      380    1.9   0.17
## Residuals 57  11395      200
```

The “Pr(>F)” column shows the p-value associated with the F statistic. In this case, the p-value is 0.17, which is greater than the significance level of 0.05. Therefore, we fail to reject the null hypothesis and conclude that tree type does not influence volume. A t-test can be thought of as a special case of ANOVA, where there are only two groups being compared. In this case, we have two groups (Beech and Oak), so we could perform a two-sample t-test to compare their mean volumes.

```
##
## Welch Two Sample t-test
##
## data: beech and oak
## t = -1, df = 53, p-value = 0.2
## alternative hypothesis: true difference in means is not equal to 0
## 95 percent confidence interval:
##  -12.33  2.17
## sample estimates:
## mean of x mean of y
##    30.2    35.2
```

The null hypothesis is that the means of the two groups are equal, and the alternative hypothesis is that they are not equal and since p-value is 0.2, there is not enough evidence to reject the null hypothesis at a significance level of 0.05. This confirms the results of the ANOVA test. We can then estimate the volumes of the two types of trees by looking at their mean:

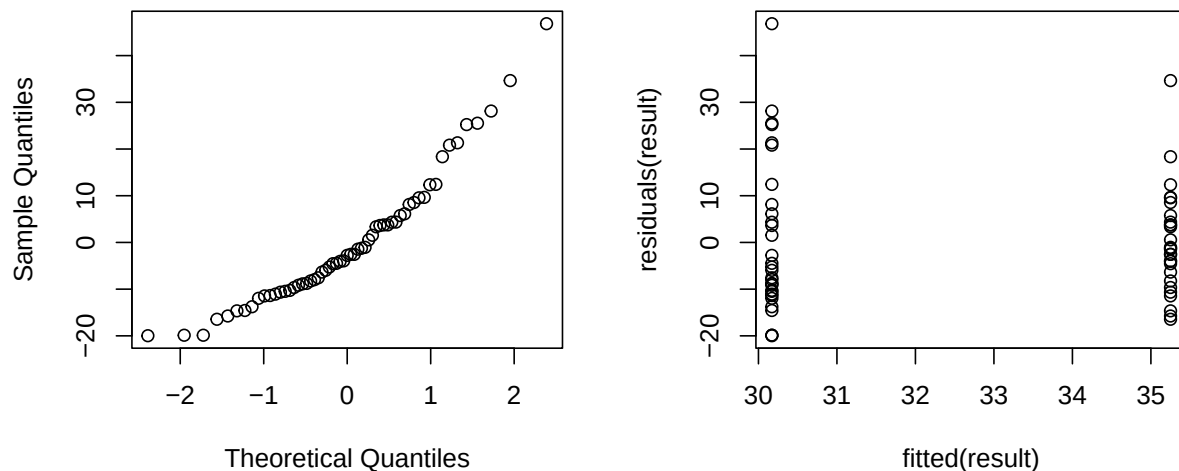
```
mean(beech) # mean volume of type beech
```

```
## [1] 30.2
```

```
mean(oak) # mean volume of type oak
```

```
## [1] 35.2
```

Lastly, we want to check whether the assumption of normality is not totally untrue using a qqnorm plot and plotting the fitted values against the residuals:



we can notice that QQ-norm does not follow a straight line, hence we can not confirm the normality assumption.

b) To investigate whether the influence of diameter and height on volume is similar for both tree types (Beech and Oak), we can fit two linear regression models.

```
#pairwise interaction between type and diameter
mod1 <- lm(volume~height+type*diameter, data=data)
anova(mod1)
```

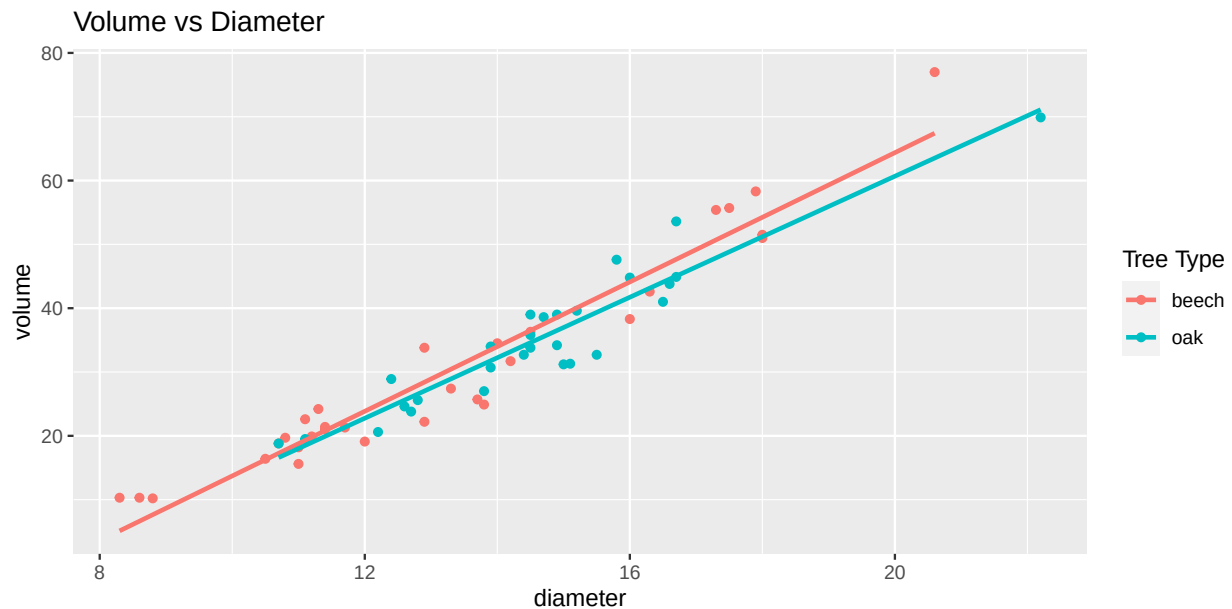
```
## Analysis of Variance Table
##
## Response: volume
##           Df Sum Sq Mean Sq F value    Pr(>F)
## height      1   2188    2188  206.21 < 2e-16 ***
## type         1    431     431   40.65 4.2e-08 ***
## diameter     1   8577    8577  808.49 < 2e-16 ***
## type:diameter 1     6        6    0.52  0.47
## Residuals   54    573        11
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
#pairwise interaction between type and height
mod2 <- lm(volume~diameter+type*height, data=data)
anova(mod2)
```

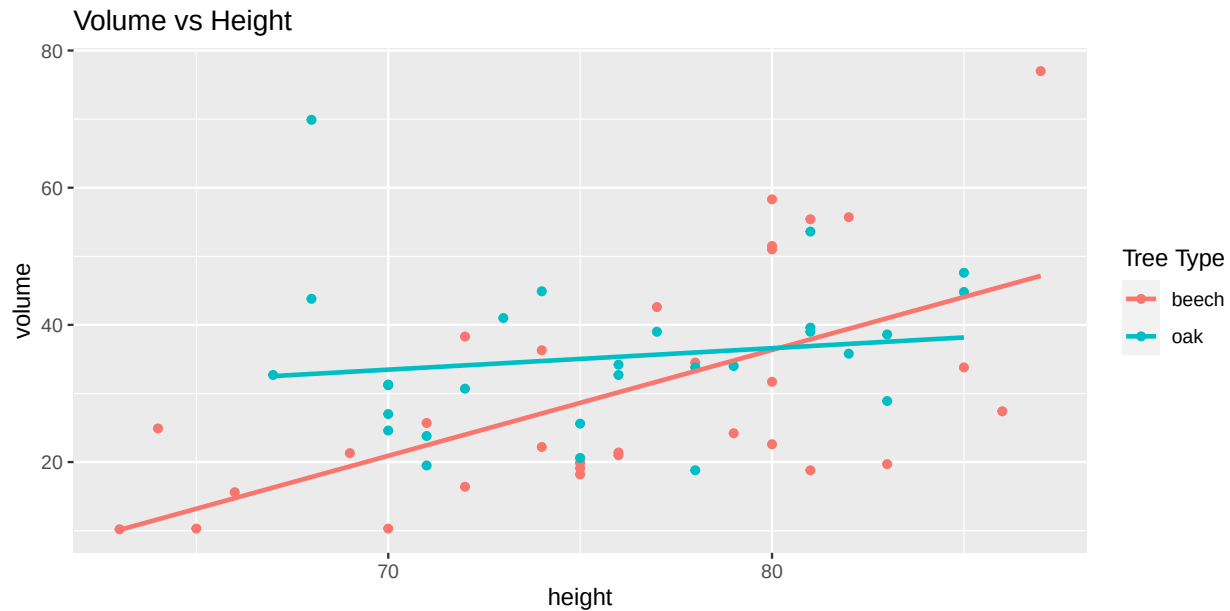
```
## Analysis of Variance Table
##
## Response: volume
##          Df Sum Sq Mean Sq F value    Pr(>F)
## diameter   1  10827   10827  1045.97 < 2e-16 ***
## type        1     45      45    4.37   0.041 *
## height      1    324     324   31.32 7.5e-07 ***
## type:height 1     19      19    1.88   0.176
## Residuals  54    559      10
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Looking at the p-values, from both models we do not reject the null hypothesis and conclude that the effect of diameter and height on volume is similar for both tree types. We want now to try and visualize the relationship between volume and diameter/height for each tree type:

```
## 'geom_smooth()' using formula = 'y ~ x'
```



```
## 'geom_smooth()' using formula = 'y ~ x'
```



From the plots, it seems like there may be a different relationship between volume and diameter/height for Beech and Oak trees. It appears to exist a strong relationship between volume and diameter for both trees. Similarly, also between volume and height we can see a significant relationship, in this case mostly for beech trees. Let's now fit two more linear regression models and compare their coefficients to confirm these observations:

```
# Linear regression model for volume vs diameter for each tree type
beechModel <- lm(volume ~ diameter, data = data, subset = type == "beech")
oakModel <- lm(volume ~ diameter, data = data, subset = type == "oak")
```

```
# Compare the coefficients
summary(beechModel)$coefficients
```

```
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -36.94      3.365   -11.0 7.62e-12
## diameter         5.07      0.247    20.5 8.64e-19
```

```
summary(oakModel)$coefficients
```

```
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -34.13      4.89    -6.98 2.04e-07
## diameter        4.74      0.33    14.35 7.24e-14
```

```
# Linear regression model for volume vs height for each tree type
beechModel2 <- lm(volume ~ height, data = data, subset = type == "beech")
oakModel2 <- lm(volume ~ height, data = data, subset = type == "oak")
```

```
# Compare the coefficients
summary(beechModel2)$coefficients
```

```
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -87.12     29.273   -2.98 0.005835
## height         1.54      0.384    4.02 0.000378
```

```
summary(oakModel2)$coefficients
```

```
##           Estimate Std. Error t value Pr(>|t|)
## (Intercept)  11.576     29.431   0.393   0.697
## height       0.313      0.388   0.806   0.427
```

c) Based on the linear regression models, we can see that both diameter and height have a positive influence on volume for both tree types. We already saw that type does not influence the volume. The coefficients for diameter are similar for both tree types, with beech having a slightly higher coefficient (5.07) compared to oak (4.74). This suggests that diameter has a similar influence on volume for both beech and oak trees. The coefficients for height are also positive for both tree types, but the magnitudes are different. The beech trees' coefficient (1.54) is much higher than that for oak trees (0.31), indicating that height has a stronger influence on volume for beech trees compared to oak trees. We can also notice it from the previous graph. Overall, we see that increasing both height and diameter result in an increase in volume, especially diameter for both type of trees. To predict the volume for a tree with the overall average diameter and height, we need to calculate the mean values for these variables:

```
##      1
## 33.5
```

```
##      1
## 31.8
```

d) Since we know that the formula to calculate the volume of a cylinder is $v = \frac{(\pi * d^2 * h)}{4}$, we simplify and substitute diameter with $diameter^2$ in the model. We also remove type in the model, since we already understood that is not significant.

```
# previous model
mod3=lm(volume~diameter+height,data=data)
summary(mod3)
```

```
##
## Call:
## lm(formula = volume ~ diameter + height, data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -6.724 -2.278 -0.034  1.820  8.629
##
## Coefficients:
##           Estimate Std. Error t value Pr(>|t|)
## (Intercept) -64.3697     5.5577  -11.58 < 2e-16 ***
## diameter      4.6325     0.1602   28.92 < 2e-16 ***
## height       0.4289     0.0755    5.68 5.1e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 3.28 on 56 degrees of freedom
## Multiple R-squared:  0.949, Adjusted R-squared:  0.947
## F-statistic: 520 on 2 and 56 DF, p-value: <2e-16
```

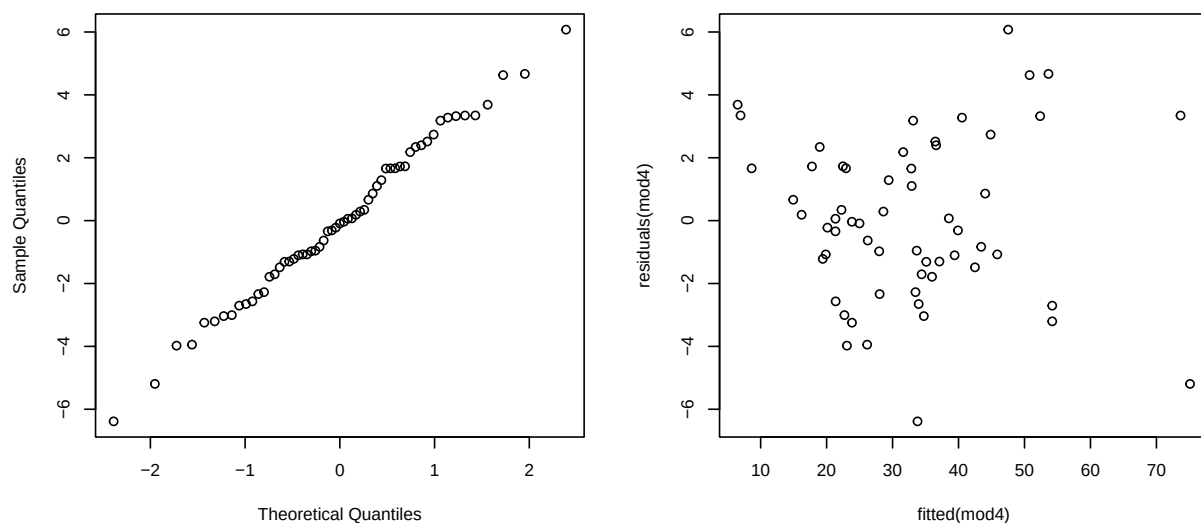
```

# new model
data$diam_sq=data$diameter^2
mod4=lm(volume~diam_sq+height,data=data)
summary(mod4)

##
## Call:
## lm(formula = volume ~ diam_sq + height, data = data)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -6.385 -1.597 -0.089  1.725  6.075
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -37.17079    4.42928  -8.39  1.8e-11 ***
## diam_sq      0.15911    0.00436  36.53 < 2e-16 ***
## height       0.49781    0.05997   8.30  2.5e-11 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 2.63 on 56 degrees of freedom
## Multiple R-squared:  0.967, Adjusted R-squared:  0.966
## F-statistic: 826 on 2 and 56 DF, p-value: <2e-16

```

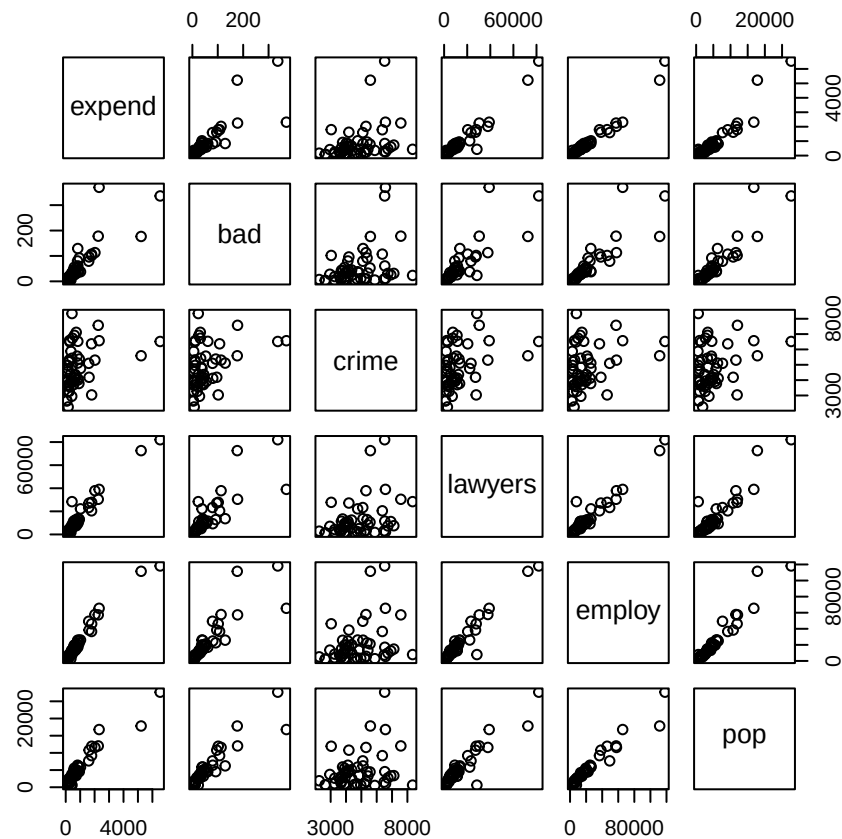
We can confirm that the model with the transformed explanatory variable is better. This is indicated by the higher F-value and lower p-value for the `diam_sq` variable compared to the `diameter` variable in the first model. Additionally, the multiple R-squared and adjusted R-squared values are higher in the second model, indicating a better fit. Lastly, we want to check the assumption of normality to understand if we can make valid assumptions from this model:



We can see that the norm qq-plot follows a straight line and the fitted-residual model does not show a pattern. This means that the normality is checked and we can draw valid conclusions from the model.

Exercise 2. Expenditure on criminal activities

a) In order to check the pairwise interactions between all the variables of interest, we can use scatter plots:

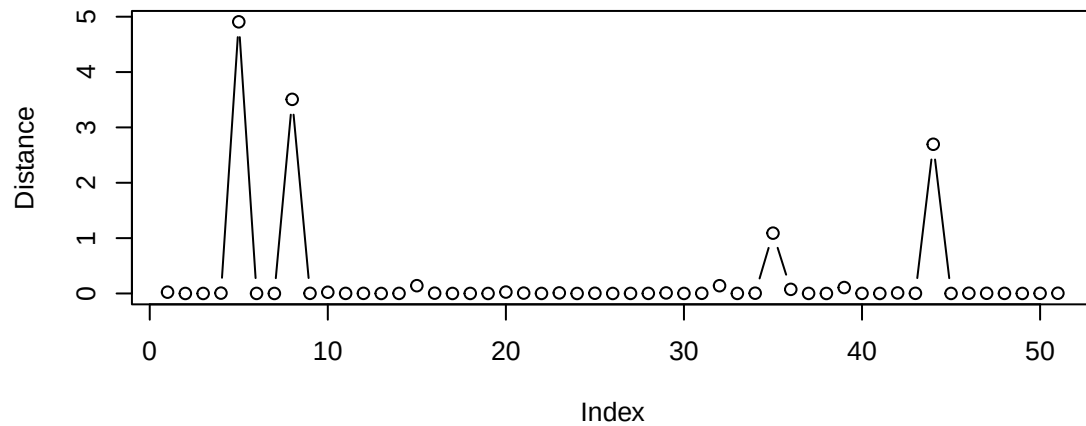


After examining the pairs plot, it is evident that there are linear relationships among the predictors, namely *bad*, *lawyer*, *employ*, and *pop*, which can lead to the issue of collinearity. Moreover, a strong positive correlation is observed between the response variable *expend* and the aforementioned predictors, except for *crime*. However, there are some outliers present in the higher range of the data, which can potentially impact the model's accuracy. The existence of collinearity among the predictors is a concern as it can make it challenging to estimate the regression coefficients due to redundant information.

In order to find influence points, we can use Cook's distance, where a distance greater than 1 indicates an outlier:

```
crimelm <- lm(expend~bad+crime+lawyers+employ+pop, data=crimedf)
```

Cook's distance for expenscrime



```
cooks.distance(crimelm)[cooks.distance(crimelm) > 1]
```

```
##      5      8     35     44
## 4.91 3.51 1.09 2.70
```

Indices 5, 8, 35 and 44 are thus regarded as influence points and can be removed from the data:

```
Updcrimedf <- crimedf[-c(5,8,35,44),]
```

Let's now investigate collinearity by examining the correlations between all the explanatory variables:

Table 1: Correlation coefficients

	bad	crime	lawyers	employ	pop
bad	1.00	0.37	0.83	0.87	0.92
crime	0.37	1.00	0.38	0.31	0.28
lawyers	0.83	0.38	1.00	0.97	0.93
employ	0.87	0.31	0.97	1.00	0.97
pop	0.92	0.28	0.93	0.97	1.00

The correlation table confirms what was observed in the pairs plot; there are strong correlations between the explanatory variables, excluding crime. The lowest correlation coefficient is 0.28, between bad and lawyers, indicating that all predictors (excluding crime) contain comparable information.

We can also use the VIF to see which variables are collinear, where a value >5 is cause for concern:

```
## Loading required package: carData
```

```
vif(lm(expend~bad+crime+lawyers+employ+pop, data=crimedf))
```

```
##      bad      crime lawyers  employ      pop
## 8.36      1.49     16.97    33.59    32.94
```

This result also supports that the variables *bad*, *lawyers*, *employ* and *pop* are collinear.

b) From now on, we will proceed without the influence points.

```
# level 1
mbad <- lm(expend~bad, data=Updcrimedf); mcrime <- lm(expend~crime, data=Updcrimedf)
mlawyers <- lm(expend~lawyers, data=Updcrimedf); memploy <- lm(expend~employ, data=Updcrimedf)
mpop <- lm(expend~pop, data=Updcrimedf)
l1m <- list(mbad, mcrime, memploy, mlawyers, mpop);
var1 <- c("bad", "crime", "employ", "lawyears", "pop")
Rsquared1 <- c()
for(m1 in l1m){
  r_squared <- summary(m1)$r.squared
  Rsquared1 <- c(Rsquared1, r_squared)
}
knitr::kable(data.frame(Predictor = var1, Rsquared = Rsquared1),
              caption = "Step-up: Level 1")
```

Table 2: Step-up: Level 1

Predictor	Rsquared
bad	0.820
crime	0.120
employ	0.956
lawyears	0.928
pop	0.931

From Table 2, it is deduced that *employ* is the variable to be included in the model. Analogously, we obtain Tables 3 and 4 which allow us to obtain the best step-up model.

Table 3: Step-up: Level 2

Predictor	r.squared	Significant
employ+bad	0.958	FALSE
employ+crime	0.965	TRUE
employ+lawyears	0.961	FALSE
employ+pop	0.961	FALSE

Table 4: Step-up: Level 3

Predictor	r.squared	Significant
employ+crime+bad	0.965	FALSE
employ+crime+lawyers	0.970	TRUE
employ+crime+pop	0.974	TRUE

The variables added in order were *employ*, *crime* and *pop*, after which no further added variables had significant p-values (notice that we omitted the last level). Hence the final model is as follows:

```
mstepup <- lm(expend~employ+crime+pop, data=Updcrimedf)
summary(mstepup)
```

```
##
## Call:
## lm(formula = expend ~ employ + crime + pop, data = Updcrimedf)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -179.99  -49.64    0.48   51.19  266.63
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -2.47e+02   5.47e+01  -4.52  4.8e-05 ***
## employ       2.09e-02   3.95e-03   5.30  3.7e-06 ***
## crime        5.43e-02   1.13e-02   4.82  1.8e-05 ***
## pop          7.14e-02   1.79e-02   4.00  0.00025 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 91.4 on 43 degrees of freedom
## Multiple R-squared:  0.974, Adjusted R-squared:  0.973
## F-statistic:  547 on 3 and 43 DF,  p-value: <2e-16
```

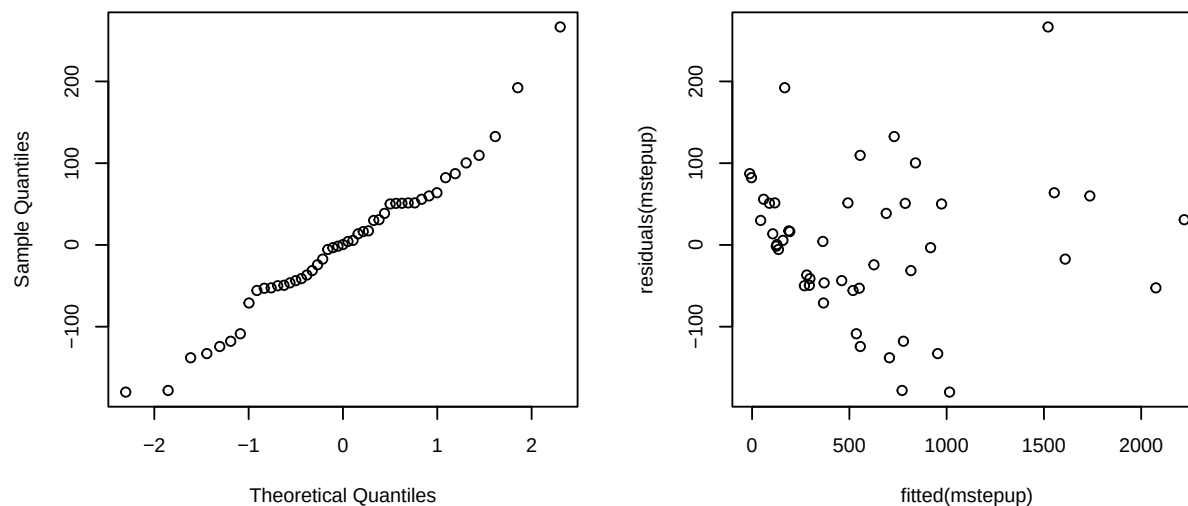
Step-up final model: $\text{expend} = -247 + 0.0209 \cdot \text{employ} + 0.0543 \cdot \text{crime} + 0.0714 \cdot \text{pop} \pm \text{error}$, with $R^2 = 0.974$.
We can check VIF for further insight:

```
vif(mstepup)
```

```
## employ  crime    pop
## 18.08   1.14  17.51
```

Although the step-up method naturally removes collinearity and produced a better model than the one obtained in *a*), which had an R-squared value of 0.957, there is still a high VIF value for *employ* and *pop*.

Finally, we check the model assumptions:



We can assume normality, although it is worth addressing that the qq plot makes small curved groups of points.

c) Using the step-up model from the previous case, the 95% prediction interval for *expend* is given by:

```
newdata <- data.frame(bad=50, crime=5000, lawyers=5000, employ=5000, pop=5000)
predict(mstepup, newdata, interval="prediction")
```

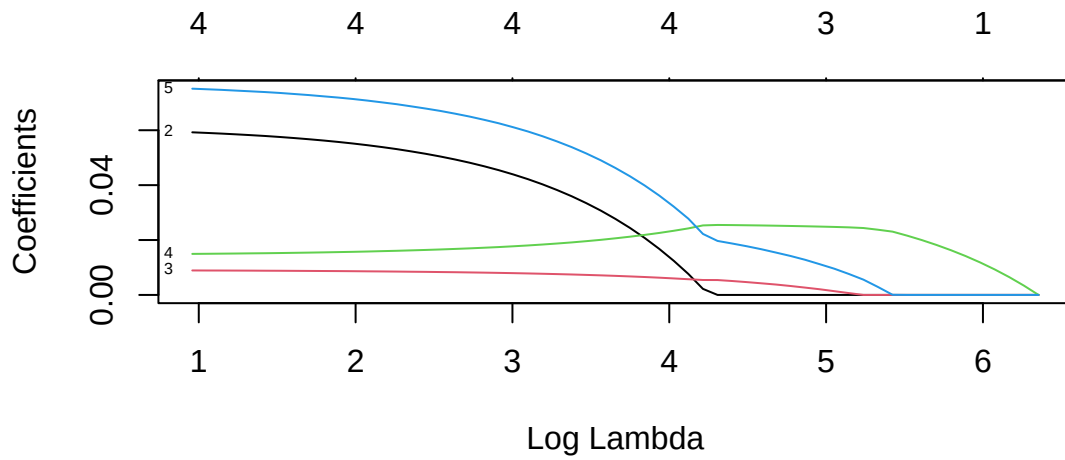
```
## fit lwr upr
## 1 486 258 713
```

One way to improve this interval would be by transforming the existing variables to better capture their relationship with the response variable. However, this would require additional analysis and experimentation. Moreover, we already removed the outliers and the inclusion of more variables in the model is not recommended, since we already tested that there are no significant variables to add.

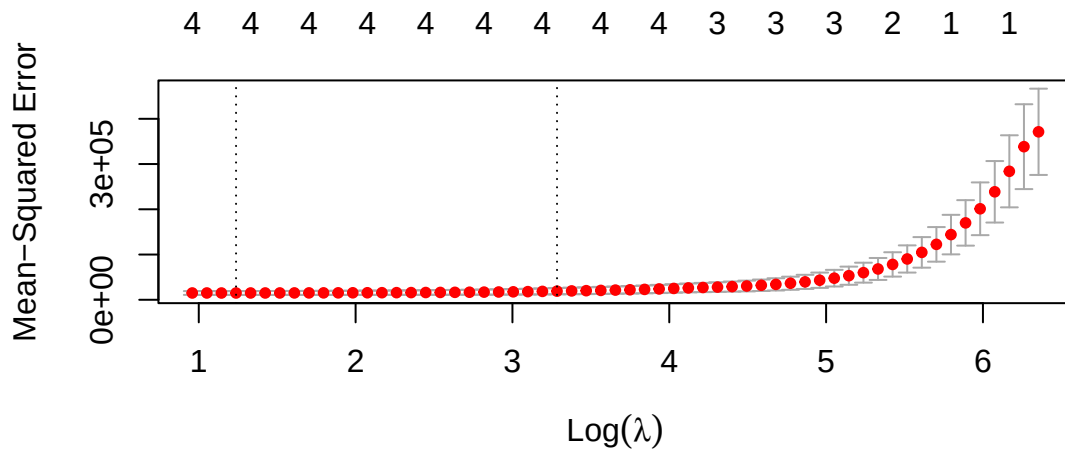
d) Let's apply the LASSO method:

```
x <- as.matrix(Updcrimedf[, exp_vars])
y <- as.matrix(Updcrimedf[, c(response)])
train <- (sample(1:nrow(x), 0.67*nrow(x))) # train by using 2/3 of the data
x.train <- x[train,]; y.train <- y[train] # data to train
x.test <- x[-train,]; y.test <- y[-train] # data to test the prediction quality
mlasso <- glmnet(x.train, y.train, alpha=1)
cvlasso <- cv.glmnet(x.train, y.train, alpha=1, type.measure='mse')
```

```
plot(mlasso, label=T, xvar="lambda") # visualization of the lasso path
```



```
plot(cvlasso) # the best lambda by cross-validation
```



```
(lambda.1se <- cvlasso$lambda.1se) # lambda for most regularized model
```

```
## [1] 26.7
```

Let's now asses the performance of the model built using Lasso regression.

```
assess.glmnet(mlasso, newx = x.test, newy = y.test, s=cvlasso$lambda.1se)
```

```
## $mse
##    s1
## 4374
```

```
## attr("measure")
## [1] "Mean-Squared Error"
##
## $mae
## s1
## 51
## attr("measure")
## [1] "Mean Absolute Error"
```

In order to compare this model to the step-up model from (b), we find the MSE for the latter:

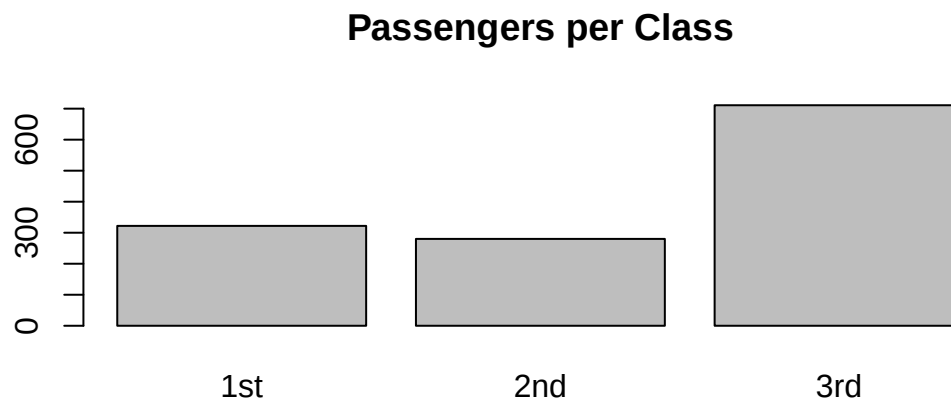
```
newdata <- data.frame(x.test)
y.pred <- predict(mstepup, newdata, interval="confidence", level=0.95)
msestepup <- mean((y.test - y.pred)^2); msestepup
```

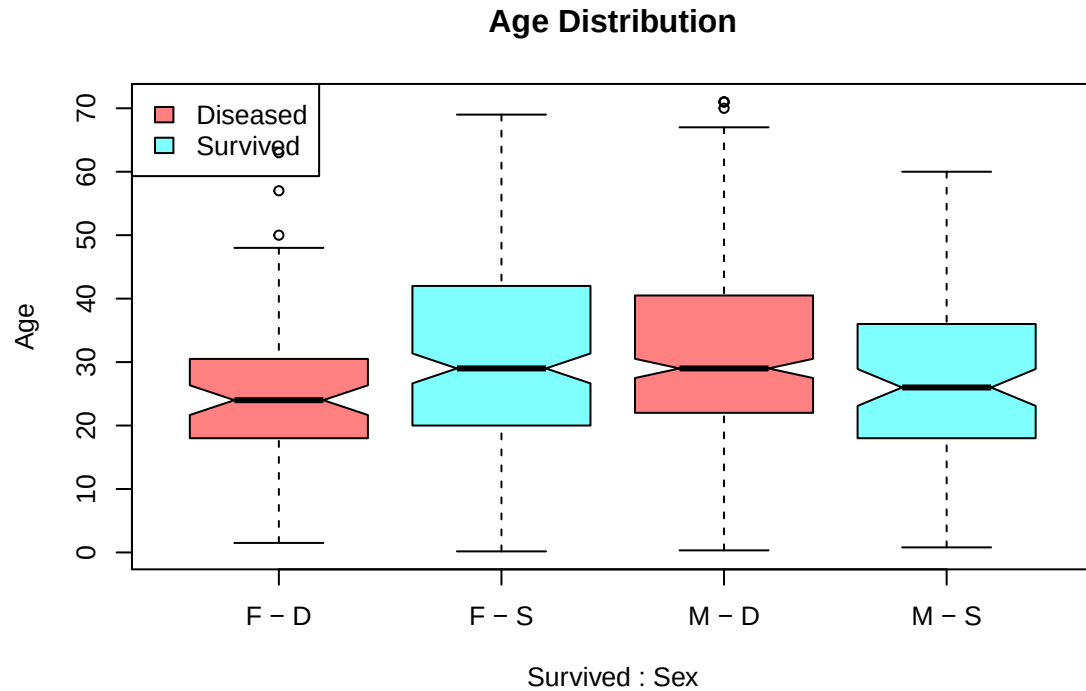
```
## [1] 6888
```

After several runs (of the Lasso method), we can conclude that the step-up model outperforms it consistently in this case (20 out of 20 tries, but we did not keep track), producing a smaller MSE. This could happen because LASSO is better suited to situations with many more explanatory variables.

Exercise 3. Titanic

a) To get a better understanding we visualized several elements from the data. First, we were interested in the number of people per class. It appears that the third class is relatively large and the first and second class are about equal size. Then we visualized the effect of age and sex on survival rate. The figure shows that the average age of surviving men is lower than the average age of the men that did not survive, this indicates a positive effect of age survival rate. However, females show the opposite effect, disputing the previous indication. Some of the data points missed the age column, these values were replaced with the average age.





A logistic regression model was created to investigate the association between the survival status and the predictors *PClass*, *Age* and *Sex*, according to code in Listing 6. The estimated odds is calculated as follows:

$$\hat{o}_k = \frac{Pr(Y_k = 1)}{Pr(Y_k = 0)} = \exp\{3.76 - 1.29Class2_k - 2.52Class3_k - 0.039177Age_k - 2.63SexMale_k\}$$

First of all, we see that age does not strongly effect the survival rate. Next, we observe a relatively high negative value for being a third class passenger, this indicates that traveling in third class negatively effects survival rate. The parameter for second class passengers also shows a negative influence on the survival rate, but the effect is not as strong as for the third class passengers. Finally, we see that being a male also has a strong negative effect on the probability of survival.

```
titanicmodel = glm(Survived ~ PClass + Age + Sex, data=titanicdata, family=binomial)
summary(titanicmodel)
```

```
##
## Call:
## glm(formula = Survived ~ PClass + Age + Sex, family = binomial,
##      data = titanicdata)
##
## Deviance Residuals:
##      Min       1Q   Median       3Q      Max
## -2.723  -0.707  -0.392   0.649   2.529
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)   3.75966    0.39757   9.46 < 2e-16 ***
## PClass2nd    -1.29196    0.26008  -4.97 6.8e-07 ***
## PClass3rd    -2.52142    0.27666  -9.11 < 2e-16 ***
```

```
## Age          -0.03918    0.00762   -5.14  2.7e-07 ***
## Sexmale      -2.63136    0.20151  -13.06  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
## Null deviance: 1025.57 on 755 degrees of freedom
## Residual deviance: 695.14 on 751 degrees of freedom
## (557 observations deleted due to missingness)
## AIC: 705.1
##
## Number of Fisher Scoring iterations: 5
```

b) The following different models were used to investigate the interaction between different predictors. All models use a null hypothesis $H_0 : \beta_{Age} = \beta_{Class}$ and a significance level $\alpha = 0.05$.

```
modelAgePClass = glm(Survived ~ Age * PClass, data=titanicdata, family=binomial)
anova(modelAgePClass, test="Chisq")
```

```
## Analysis of Deviance Table
##
## Model: binomial, link: logit
##
## Response: Survived
##
## Terms added sequentially (first to last)
##
##
##          Df Deviance Resid. Df Resid. Dev Pr(>Chi)
## NULL                    755      1026
## Age           1         2.8      754      1023   0.091 .
## PClass        2       112.8      752        910 <2e-16 ***
## Age:PClass    2         1.2      750        909   0.558
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
modelAgeSex = glm(Survived ~ Age * Sex, data=titanicdata, family=binomial)
anova(modelAgeSex, test="Chisq")
```

```
## Analysis of Deviance Table
##
## Model: binomial, link: logit
##
## Response: Survived
##
## Terms added sequentially (first to last)
##
##
##          Df Deviance Resid. Df Resid. Dev Pr(>Chi)
## NULL                    755      1026
## Age           1         2.8      754      1023   0.091 .
```

```
## Sex      1      227.1      753      796 < 2e-16 ***
## Age:Sex  1      25.0      752      771 5.6e-07 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The first model, which models the interaction between *Age* and *Class*, returns a p-value of 0.558. This does not reject H_0 , meaning that there is no interaction between predictor *Age* and factor *Class*. The second model on the interaction between *Age* and *Sex* returns $p < 0.001$. This value rejects our H_0 , which means that the two predictors interact.

From this point, we work with a model that uses the *PClass* predictor, but it also integrates the fact that *Age* and *Sex* interact.

```
modelFinal = glm(Survived ~ PClass + Age * PClass, data=titanicdata, family=binomial)
anova(modelFinal, test="Chisq")
```

```
## Analysis of Deviance Table
##
## Model: binomial, link: logit
##
## Response: Survived
##
## Terms added sequentially (first to last)
##
##
##          Df Deviance Resid. Df Resid. Dev Pr(>Chi)
## NULL                      755      1026
## PClass      2      78.0      753      948 < 2e-16 ***
## Age         1      37.6      752      910 8.6e-10 ***
## PClass:Age  2       1.2      750      909   0.56
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The estimate of the probability of survival for each combination of levels of the factors *Class* and *Sex* for a person of age 53 can be found in the following table.

Table 5: Probability of survival for each combination of levels of the factors *Class* and *Sex* for a person of age 53

	Male	Female
Class 1	0.5358867	0.7119470
Class 2	0.5115599	0.6754746
Class 3	0.5035067	0.6097319

```
titanicdata$PClass = factor(titanicdata$PClass)
titanicdata$Sex = factor(titanicdata$Sex)

summary(modelFinal)
```

```
##
## Call:
```

```
## glm(formula = Survived ~ PClass + Age * PClass, family = binomial,
##      data = titanicdata)
##
## Deviance Residuals:
##      Min       1Q   Median       3Q      Max
## -1.999  -0.909  -0.669   1.075   2.202
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)    1.92298    0.43625     4.41  1.0e-05 ***
## PClass2nd      -0.74428    0.57155    -1.30  0.19284
## PClass3rd      -2.29007    0.54057    -4.24  2.3e-05 ***
## Age            -0.03584    0.00996    -3.60  0.00032 ***
## PClass2nd:Age  -0.01321    0.01587    -0.83  0.40519
## PClass3rd:Age   0.00464    0.01594     0.29  0.77090
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 1025.57  on 755  degrees of freedom
## Residual deviance:  908.75  on 750  degrees of freedom
## (557 observations deleted due to missingness)
## AIC: 920.8
##
## Number of Fisher Scoring iterations: 4
```

```
m_names = list(c("Class1", "Class2", "Class3"), c("Male", "Female"))
results = matrix(, nrow=3, ncol=2, dimnames=m_names)

classLabels = c("1st", "2nd", "3rd")
genderLabels = c("male", "female")
for (i in seq_along(classLabels)) {
  for (j in seq_along(genderLabels)) {
    newData = data.frame(Age=53, Sex=genderLabels[j], PClass=classLabels[i])
    x = predict(modelFinal, newData, type="response")
    results[i, j] = 1/(1 + exp(-x))
  }
}
```

c) The second approach is not wrong, contingency tables should produce the similar conclusions as the logistic regression model. The logistic regression model is a rather complex model, it does not only test for predictor interaction, but it also creates a model which can be used for prediction. So the logistic regression model is more versatile than the contingency table. The complexity of the model can also be considered as a disadvantage, as it is less intuitive to interpret.

We propose to split the data into a training and a test set to assess the performance of our predictor. The algorithm can use the training set to train the model, i.e. it tries to find the optimal parameters. The test set contains data points which the model has never seen before, these data points are used to assess the performance of the model. If the model is able to accurately predict the survival on unseen data, then we can conclude that the model generalizes the data well.

d) We created contingency tables to investigate whether factor passenger class has an effect on the survival status, see Table 6. The built-in function `chisq.test(classTable)` shows a result of $p < 0.001$, indicating that passenger class and survival are dependent.

Table 6: Number of passengers per class against survival

	Class 1	Class 2	Class 3
Survived	193	119	138
Deceased	129	161	573

A similar thing was done for the factor gender, see Table 7. Because this is a 2x2 table, we can use the Fisher's exact test. This is a built-in function in R, `fisher.test(genderTable)`, it also reports a result of $p < 0.001$, thus the gender and survival rate are also dependent.

Table 7: Number of passengers per gender against survival

	Male	Female
Survived	142	308
Deceased	709	154

e) The second approach is not wrong, contingency tables should produce the similar conclusions as the logistic regression model. The logistic regression model is a rather complex model, it does not only test for predictor interaction, but it also creates a model which can be used for prediction. So the logistic regression model is more versatile than the contingency table. The complexity of the model can also be considered as a disadvantage, as it is less intuitive to interpret.

Exercise 4. Military coups

a) All variables of the coups data set were loaded in a Poisson GLM, using the following procedure.

```
coupsdata=read.table(file="Data/coups.txt", header=TRUE)
coupsglm=glm(miltcoup ~ oligarchy + pollib + parties +
              pctvote + popn + size + numelec + numregim,
              family=poisson,
              data=coupsdata )
summary(coupsglm)

##
## Call:
## glm(formula = miltcoup ~ oligarchy + pollib + parties + pctvote +
##      popn + size + numelec + numregim, family = poisson, data = coupsdata)
##
## Deviance Residuals:
##      Min       1Q   Median       3Q      Max
## -1.344  -0.954  -0.259   0.391   1.695
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept) -0.510269   0.905330  -0.56   0.5730
## oligarchy    0.073081   0.034596   2.11   0.0346 *
## pollib      -0.712978   0.272563  -2.62   0.0089 **
## parties      0.030774   0.011187   2.75   0.0059 **
## pctvote      0.013872   0.009753   1.42   0.1549
## popn         0.009343   0.006595   1.42   0.1566
```

```
## size          -0.000190   0.000248   -0.76   0.4445
## numelec       -0.016078   0.065484   -0.25   0.8060
## numregim      0.191735   0.229289    0.84   0.4030
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for poisson family taken to be 1)
##
##      Null deviance: 65.945  on 35  degrees of freedom
## Residual deviance: 28.668  on 27  degrees of freedom
## AIC: 111.5
##
## Number of Fisher Scoring iterations: 6
```

These results indicate that the military coups are significantly predicted by 3 factors: *oligarchy*, *pollib* and *parties*.

1. The number of years the country is ruled by a military oligarchy has a positive effect on the number of military coups; an increase in years of military oligarchy predicts an increase in military coups.
2. The political liberalization level has a negative effect on the number of military coups; the fewer civil rights a country has, the number of coups increases.
3. The number of political parties has a positive effect on the number of military coups; as such that when a country has more political parties, a greater number of military coups can be expected.

All other variables in the model didn't have a significant effect on the number of military coups.

b) The number of exploratory variables in the former Poisson regression was reduced by removing the variable that has the highest P-value in the model. This procedure was repeated until all variables in the model are significant. The following variables were successively removed from the model: *numelec*, *numregim*, *size*, *popn*, *pctvote*. The final model contains the following explanatory variables: *oligarchy*, *pollib*, *parties*.

```
coupsglm5 <- step(coupsglm, direction = "backward", trace = FALSE, family = "poisson")
summary(coupsglm5)
```

```
##
## Call:
## glm(formula = miltcoup ~ oligarchy + pollib + parties, family = poisson,
##      data = coupsdata)
##
## Deviance Residuals:
##      Min       1Q   Median       3Q      Max
## -1.358   -1.042   -0.286    0.628    1.752
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  0.25138   0.37269   0.67   0.500
## oligarchy    0.09262   0.02178   4.25 2.1e-05 ***
## pollib      -0.57410   0.20438  -2.81  0.005 **
## parties      0.02206   0.00896   2.46  0.014 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
```

```
## (Dispersion parameter for poisson family taken to be 1)
##
##      Null deviance: 65.945  on 35  degrees of freedom
## Residual deviance: 32.856  on 32  degrees of freedom
## AIC: 105.7
##
## Number of Fisher Scoring iterations: 5
```

While in the full model *oligarchy* was barely significant with a value of $p = 0.034$, in the reduced model this variable has a very small p-value ($p < 0.001$), which indicates that the presence of an oligarchy is a valid predictor of a military coup. The predictive value of the number of political parties in a country reduced in significance. The political liberalization level remained similar in significance.

c) Using the model from the previous section, the number of coups can be predict for each of the levels of political liberalization. For the variables *oligarchy* and *parties* averages were calculated and fed into the model, respectively 5.222 and 17.083.

```
# Level 0 No civil rights for political expression
pollib0 = data.frame(
  pollib=0,
  oligarchy=mean(coupsdata[, 'oligarchy']),
  parties=mean(coupsdata[, 'parties'])
)
predict(coupsglm5, newdata=pollib0, type="response")
```

```
##      1
## 3.04
```

```
# Level 1 Limited civil rights for expression
pollib1 = data.frame(
  pollib=1,
  oligarchy=mean(coupsdata[, 'oligarchy']),
  parties=mean(coupsdata[, 'parties'])
)
predict(coupsglm5, newdata=pollib1, type="response")
```

```
##      1
## 1.71
```

```
# Level 2 Full civil rights
pollib2 = data.frame(
  pollib=2,
  oligarchy=mean(coupsdata[, 'oligarchy']),
  parties=mean(coupsdata[, 'parties'])
)
predict(coupsglm5, newdata=pollib2, type="response")
```

```
##      1
## 0.964
```

From these predictions can be concluded that more military coups are expected if less civil rights are granted in the average African country. When no civil rights exist, the predicted number of coups is a little over 3, while with full civil rights, the predicted number is just a little under 1.